

Transformation in ETL

Question 1: Define Data Transformation in ETL and explain why it is important.

Answer: **Data Transformation** in ETL refers to the process of converting raw data from its original format into a clean, structured and usable form. This step occurs after and before loading into the target system.

Transformation includes:

Standardization: Converts units, formats or currencies into common standard.

Data Mapping: Aligning codes or column names (**Ex – Cust_id to CustomerID**).

Derivation: Creates new fields from existing ones (**Ex – Profit = Revenue – Cost**).

Cleaning: Handling missing values, inconsistencies and duplicates.

Encoding: Converting categorical values into numeric form for analysis.

Aggregation: Summarizing data (**Ex – Total Sales Per Region**).

Data Transformation's Importance:

1. **Enable Integration:** Aligns mismatched schemas (like column names or codes) so data from multiple sources can be combined without any interruption.
2. **Improves Data Quality:** Removes Duplicates, handles missing data, fixes errors and making reliable for analysis.
3. **Prepare data for Machine Learning:** Encoding categorical variables and normalizing numeric values ensures models can process data effectively.
4. **Reduces Bias & Errors:** Proper handling of missing values to prevent skewed results.
5. **Ensures Consistency:** Data from different sources varied in formats (like dates as DD/MM/YYYY vs MM/DD/YYYY). Transformation standardizes them.

Question 2: List any four common activities involved in Data Cleaning.

Answer: Data cleaning is a crucial step in ETL because it ensures accuracy, consistency, and reliability of the dataset. Here are **four common activities** involved:

1. **Handling Missing Values**
 - >> Techniques: deletion, imputation (mean/median/mode), forward/backward fill, or flagging.
 - >> Example: Filling missing ages in a customer dataset with the median value.
2. **Removing Duplicates**
 - >> Ensures that repeated records don't distort analysis.
 - >> Example: Dropping duplicate rows where the same customer ID appears multiple times.
3. **Standardizing Formats**
 - >> Converting inconsistent formats into a uniform structure.
 - >> Example: Dates (DD/MM/YYYY vs. MM-DD-YYYY) or currencies (USD vs. INR).
4. **Correcting Inconsistencies**
 - >> Fixing mismatched or inconsistent categories and values.
 - >> Example: Standardizing "NY" and "New York" to a single category.

Question 3: What is the difference between Normalization and Standardization?

Answer: There are four major criteria between Normalization and Standardization.

Range: Normalization has fixed range (like 0 to 1, -1 to 1) whether Standardization has no fixed range.

Focus: Normalization contain scale values uniformly but Standardization contain Centre data like (mean = 0, SD = 1).

Best for: Normalization uses Magnitude-sensitive models and Standardization uses Distribution-sensitive models.

Example: Rescaling exam scores in Normalization and Z-score for statistical analysis in Standardization.

Question 4: A dataset has missing values in the “Age” column. Suggest two techniques to handle this and explain when they should be used.

Answer: Two common techniques are:

1. Median Imputation

>> **How it works:** Replace missing ages with the median value of the “Age” column.

>> **When to use:**

1. Best for skewed numeric data (like age, income, or medical costs).
2. Median is resistant to outliers, so extreme values won’t distort the imputation.

>> **Example:** If most ages are between 20–40 but a few are 90+, the median (say 28) is a fair replacement.

2. Flagging Missing Values

>> **How it works:** Create a new binary column (e.g., Age_missing) where 1 indicates missing and 0 indicates present.

>> **When to use:**

1. Best when missingness itself carries meaning (e.g., users who skip age may behave differently).
2. Preserves information about missingness for downstream models.

>> **Example:** In a survey dataset, people who don’t disclose age might represent a specific demographic pattern.

Question 5: Convert the following inconsistent “Gender” entries into a standardized format (“Male”, “Female”): ["M", "male", "F", "Female", "MALE", "f"]

Answer:

Given inconsistent entries: ["M", "male", "F", "Female", "MALE", "f"]

We want to convert them into a standardized format: “Male” and “Female”.

>> Convert **all values to lowercase** → ["m", "male", "f", "female", "male", "f"]

>> **Map values to standardized labels:**

>> "m" or "male" → "Male"
>> "f" or "female" → "Female"

Result: ["Male", "Male", "Female", "Female", "Male", "Female"]

Question 6: What is One-Hot Encoding? Give an example with the categories: “Red, Blue, Green”.

Answer: One-Hot Encoding is a technique used in data pre-processing to convert categorical variables into a numerical format that machine learning models can understand.

>> Each category is represented as a binary vector.
>> The vector length equals the number of categories.
>> Only one element is “hot” (set to 1), while the rest are 0.

Example:

We have 3 categories → vector length = 3.

Category	Red	Blue	Green
Red	1	0	0
Blue	0	1	0
Green	0	0	1

So:

>> “Red” → [1, 0, 0]
>> “Blue” → [0, 1, 0]
>> “Green” → [0, 0, 1]

Question 7: Explain the difference between Data Integration and Data Mapping in ETL.

Answer: Both Data Integration and Data Mapping are key steps in the ETL pipeline, but they serve different purposes.

1. Data Integration

>> **Definition:** The process of combining data from multiple sources into a single, unified view.
>> **Goal:** To make diverse datasets work together seamlessly.
>> **Example:** Merging customer data from a CRM system, sales transactions from an ERP system, and web analytics data into one data warehouse.
>> **Importance:**
 >> Enables holistic analysis.
 >> Removes silos between systems.
 >> Provides a consolidated dataset for reporting and decision-making.

2. Data Mapping

>> **Definition:** The process of defining how fields from one dataset correspond to fields in another.
>> **Goal:** To align schema differences and ensure consistency.

>> **Example:** Mapping Cust_ID in one source to CustomerID in another, or mapping “M”/“F” to “Male”/“Female”.

>> **Importance:**

- >> Ensures correct alignment of attributes.
- >> Prevents mismatches during integration.
- >> Supports transformations like standardization and encoding.

3. Key Difference

Aspect	Data Integration	Data Mapping
Focus	Combining multiple datasets	Aligning fields/attributes
Scope	Broad (multi-source consolidation)	Specific (field-to-field correspondence)
Example	Merging CRM + ERP + Analytics	Mapping Cust_ID → CustomerID
Role in ETL	Happens during Extract + Load	Happens during Transform

Question 8: Explain why Z-score Standardization is preferred over Min-Max Scaling when outliers exist.

Answer: 1. Min-Max Scaling

>> **Definition:** Rescales values into a fixed range (commonly 0–1).

Problem with Outliers:

- >> Outliers drastically affect xmin and xmax.
- >> This compresses the majority of values into a very narrow range.
- >> Example: If most ages are between 20–40 but one person is 120, the scaling squeezes normal ages close together, reducing variability.

2. Z-score Standardization

>> **Definition:** Rescales data so that the mean = 0 and standard deviation = 1.

Advantage with Outliers:

- >> Outliers only affect the mean and standard deviation slightly, not the entire scale.
- >> Normal values remain spread out meaningfully.
- >> Outliers are represented as large positive or negative z-scores, making them easy to detect.